

On the connection between Wess–Zumino–Witten and Chern–Simons theories

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Received 4 June 1991

We analyze the connection between the Wess–Zumino–Witten and Chern–Simons theories from a novel viewpoint: stochastic quantization. We start from a Wess–Zumino–Witten theory in two dimensions and show that its stochastic partition function equals the BRST partition function for the Chern–Simons theory in three dimensions through an adequate interpretation of the manifolds involved. We also comment on the relation between these theories in the framework of canonical quantization for systems with constraints.

Topological quantum field theories (TQFTs) introduced by Witten [1–3] have attracted much interest in recent years not only for their mathematical relevance but also for their rich physical structure.

Connections between various kinds of TQFTs in different numbers of space–time dimensions can be established. In particular, Horowitz [4] has shown within the canonical quantization approach for systems with constraints that the three-dimensional non-abelian Chern–Simons (CS) theory [5] arises naturally from the topological Yang–Mills (TYM) theory [1] in $d=4$. This connection has also been established using a completely different approach. Indeed, it has been shown [6–8] that the partition function which arises from stochastic quantization of the CS theory in $d=3$ coincides with the partition function constructed by Witten for TYM theory in $d=4$. The stochastic quantization method can also be used to establish analogous relations between TQFTs in other dimensions [9].

Another important connection has been discovered starting from $d=3$ CS theory. Indeed, Witten has shown [3] that this model is intimately related to rational conformal field theories in $d=2$. This idea has been elaborated by Moore and Seiberg [10], Elitzur et al. [11] and Alvarez-Gaumé et al. [12], who illustrated how the Wess–Zumino–Witten (WZW) theory arises from quantization of the $d=3$ CS theory.

It is the purpose of the present paper to study the connection between $d=2$ WZW theory and $d=3$ CS theory using the stochastic quantization approach. In this way we will be able to show that the stochastic partition function for the WZW model coincides with the partition function for the CS theory, with the stochastic time providing a link between $d=2$ and $d=3$. We think that our approach sheds light on the way the connection between WZW and CS models is realized and also on the remarkable relation between different kinds of TQFTs in different space–time dimensions.

In order to clarify the way in which we will establish the connection between WZW theory in $d=2$ for a field $g(x)$ and the CS theory in $d=3$ for gauge fields $A_i(y)$ related to $g(x)$, we first consider, following Baulieu and Grossman [13], the simpler case of a model with a scalar field ϕ with an action $S^{(n)}$ defined on an n -dimensional manifold M_n and how this model is related through stochastic quantization to an $(n+1)$ -dimensional model with action $S^{(n+1)}$ given by

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$$S^{(n+1)} = \int -\frac{\delta S^{(n)}}{\delta \phi} \frac{\partial \phi}{\partial \tau} d^n x d\tau, \quad (1)$$

where τ is the stochastic time, $\tau \in [0, T]$. The manifold for this model is $M_{n+1} = M_n \times [0, T]$.

It is well known that at finite stochastic time T , the stochastic partition function for the model with action $S^{(n)}$ is given by [14]

$$\mathcal{Z}_{\text{stoch}}[S^{(n)}] = \int \mathcal{D}\phi \det \Delta \exp \left\{ -\frac{1}{2} \int \left[\left(\frac{\delta S^{(n)}}{\delta \phi} \right)^2 + \left(\frac{\partial \phi}{\partial \tau} \right)^2 \right] d^n x d\tau \right\}, \quad (2)$$

where $\det \Delta$ arises in the process of passing from the noise η integration to the ϕ integration. Δ is computed from the Langevin equation

$$\frac{\partial \phi}{\partial \tau} = -\frac{\delta S^{(n)}}{\delta \phi} + \eta, \quad (3)$$

where $\phi = \phi(x, \tau)$; it is given by the expression

$$\Delta = \frac{\partial}{\partial \tau} + \frac{\delta^2 S^{(n)}}{\delta \phi^2}. \quad (4)$$

The notation in eq. (2) is intended to make explicit that $\mathcal{Z}_{\text{stoch}}[S^{(n)}]$ corresponds to the stochastic partition function for a model with action $S^{(n)}$.

Let us now consider a quantum theory in $n+1$ dimensions defined from the classical action (1). Note that this action is in fact the integral of a total derivative. Moreover, it is connected with $S^{(n)}$ through the equation

$$S^{(n+1)} = -S^{(n)}[\phi(x, T)] + S^{(n)}[\phi(x, 0)]. \quad (5)$$

In order to compute the partition function for such an action, which only depends on the field at initial and final values of τ , one has to fix its large (topological) symmetry. In fact, under a change

$$\phi(x, \tau) \rightarrow \phi(x, \tau) + \delta \phi, \quad (6)$$

$$\delta \phi = \epsilon(x, \tau), \quad (7)$$

with $\epsilon(x, \tau)$ an arbitrary function satisfying $\epsilon(x, 0) = \epsilon(x, T) = 0$, the action $S^{(n+1)}$ is invariant. Following the usual BRST procedure, it is easy to see that the corresponding partition function takes the form

$$\mathcal{Z}_{\text{BRST}}[S^{(n+1)}] = \int \mathcal{D}\phi \mathcal{D}b \mathcal{D}\bar{\psi} \mathcal{D}\psi \exp \left[- \int \left(-\frac{\delta S^{(n)}}{\delta \phi} \frac{\partial \phi}{\partial \tau} + \{Q, \bar{\psi}(F[\phi] - \frac{1}{2}b)\} \right) d^n x d\tau \right]. \quad (8)$$

Here the linear operator $\{Q, \}$, acting on a functional V , is defined from the BRST variation of V as

$$\delta_{\text{BRST}} V = \lambda \{Q, V\}, \quad (9)$$

with λ an anticommuting parameter. We have implemented the gauge fixing of the large symmetry (7) through a gauge function F and a Lagrange multiplier b . BRST transformation are defined by

$$\{Q, \bar{\psi}\} = b, \quad \{Q, b\} = 0, \quad \{Q, \phi\} = \psi, \quad \{Q, \psi\} = 0, \quad (10)$$

where ψ and $\bar{\psi}$ are anticommuting ghost and antighost, respectively (obviously $Q^2 = 0$).

Now, if we choose as gauge function F ,

$$F[\phi] = \frac{\delta S^{(n)}}{\delta \phi} + \frac{\partial \phi}{\partial \tau}, \quad (11)$$

it is easy to see that

$$\mathcal{Z}_{\text{BRST}}[S^{(n+1)}] = \int \mathcal{D}\phi \mathcal{D}b \mathcal{D}\bar{\psi} \mathcal{D}\psi \exp \left\{ - \int \left[- \frac{\delta S^{(n)}}{\delta \phi} \frac{\partial \phi}{\partial \tau} - \frac{1}{2} b^2 + b \left(\frac{\delta S^{(n)}}{\delta \phi} + \frac{\partial \phi}{\partial \tau} \right) + \bar{\psi} \left(\frac{\partial}{\partial \tau} + \frac{\delta^2 S^{(n)}}{\delta \phi^2} \right) \psi \right] d^n x d\tau \right\}. \quad (12)$$

Integrating the auxiliary field b and ghost and antighost fields ψ and $\bar{\psi}$ it follows immediately from eqs. (2) and (12) that

$$\mathcal{Z}_{\text{stoch}}[S^{(n)}] = \mathcal{Z}_{\text{BRST}}[S^{(n+1)}]. \quad (13)$$

A connection between the stochastically quantized theory resulting from action $S^{(n)}$ and the BRST quantized theory arising from action $S^{(n+1)}$ is then established.

This connection between two models classically defined by actions $S^{(n)}$ and $S^{(n+1)}$ in n and $n+1$ space-time dimensions, respectively, can be analyzed using a completely different approach [4], namely canonical quantization for systems with constraints. To see this, let us start from action $S^{(n+1)}$ defined in eq. (1). Clearly, the momentum π conjugate to ϕ (with τ taken as physical time in $n+1$ dimensions) is (see eq. (1))

$$\pi = - \frac{\delta S^{(n)}}{\delta \dot{\phi}}. \quad (14)$$

Classically the associated constraint

$$C \equiv \pi + \frac{\delta S^{(n)}}{\delta \phi} = 0 \quad (15)$$

generates large transformations of the form (7):

$$\delta \phi = \epsilon. \quad (16)$$

At the quantum level, one has to impose this constraint through an operator acting on the wave functionals $\Psi[\phi]$ defining the quantum states

$$C\Psi[\phi] = 0. \quad (17)$$

If we now impose canonical commutation relations for ϕ and π , eq. (17) becomes

$$\frac{1}{i} \frac{\delta \Psi}{\delta \phi} + \frac{\delta S^{(n)}}{\delta \phi} \Psi = 0, \quad (18)$$

which has as unique solution

$$\Psi[\phi] = \exp(-iS^{(n)}[\phi]). \quad (19)$$

The connection between the $(n+1)$ -dimensional theory with wave functional (19) and the n -dimensional theory with action $S^{(n)}$ is again established through this result.

Let us stress that within stochastic quantization (under some general conditions) the probability distribution for the field ϕ converges for $T \rightarrow \infty$ towards the equilibrium distribution $\exp(-S^{(n)}[\phi])$, which coincides with the euclidean version of the quantum state (19). It is remarkable that the stochastic quantization approach, starting from an action $S^{(n)}$, allows one to establish a link with a topological theory with action $S^{(n+1)}$, the same link that one obtains starting from this topological action and applying the canonical quantization approach for a theory with constraints.

Let us extend the result summarized in eq. (13) to the case of interest in the present work, i.e. the connection between WZW theory and the CS model. To this end we choose $S^{(n=2)}$ as the Wess–Zumino–Witten action, $S^{(n=2)} = S_{\text{WZW}}$, in the form it takes in light-cone like coordinates:

$$S_{\text{WZW}} = \frac{k}{4\pi} \int_{\partial Y} dx dt \operatorname{Tr}(g^{-1} \partial_t g g^{-1} \partial_x g) + \frac{k}{12\pi} \int_Y d^3y \epsilon^{ijk} \operatorname{Tr}(g^{-1} \partial_i g g^{-1} \partial_j g g^{-1} \partial_k g). \quad (20)$$

We shall take $Y = D \times \mathbb{R}$, with D being a disk (of course other cases can be considered). Then ∂Y is $S^1 \times \mathbb{R}$ with $\mathbb{R}$ interpreted as time. We take real coordinates t along the time direction and x on S^1 , $g(x) = g(x + 2\pi)$. The field g is an element of $SU(N)$ and k is an integer (our conventions for the $SU(N)$ generators are $\operatorname{Tr}(t^a t^b) = \delta^{ab}$). The analogue of eq. (2) is

$$\mathcal{Z}_{\text{stoch}}^{\text{WZW}} = \int \mathcal{D}g \det \Delta \exp \left\{ -\frac{1}{2} \int_{M_3} \operatorname{Tr} \left[\left(\frac{\delta S_{\text{WZW}}}{-ig^{-1} \delta g} \right)^2 + (-ig^{-1} \partial_\tau g)^2 \right] dx dt d\tau \right\}, \quad (21)$$

where $M_3 = \partial Y \times [0, T]$. In eq. (21) $\delta S_{\text{WZW}} / -ig^{-1} \delta g$ takes the form

$$\frac{\delta S_{\text{WZW}}}{-ig^{-1} \delta g} = -\frac{ik}{2\pi} \partial_t (g^{-1} \partial_x g), \quad (22)$$

and hence the expression for Δ is

$$\Delta = \partial_\tau + [g^{-1} \partial_\tau g,] + \frac{k}{2\pi} \partial_t (\partial_x + [g^{-1} \partial_x g,]). \quad (23)$$

Following the same steps as for the scalar case, one can again prove that

$$\mathcal{Z}_{\text{stoch}}^{\text{WZW}} = \mathcal{Z}_{\text{BRST}}[S^{(n+1)}], \quad (24)$$

provided that

$$S^{(n+1)} = \int_{M_3} -\operatorname{Tr} \left(-ig^{-1} \partial_\tau g \frac{\delta S_{\text{WZW}}}{-ig^{-1} \delta g} \right) dx dt d\tau \quad (25)$$

and the gauge function is chosen as

$$F = -ig^{-1} \partial_\tau g + \frac{\delta S_{\text{WZW}}}{-ig^{-1} \delta g}. \quad (26)$$

We have then obtained, as in the scalar case, an identity between the stochastic partition function for the WZW model and the partition function for a theory with action (25) when the gauge condition is (26).

We shall now show that $S^{(n+1)}$ in eqs. (24), (25) can be associated with the CS action, or to be more precise, we shall prove that $\mathcal{Z}_{\text{BRST}}[S^{(n+1)}]$ is in fact the partition function for the $d=3$ CS theory defined on M_3 when a certain gauge condition is chosen. Let us begin writing the explicit form of $\mathcal{Z}_{\text{BRST}}[S^{(n+1)}]$ in eq. (24):

$$\begin{aligned} \mathcal{Z}_{\text{BRST}}[S^{(n+1)}] = & \int \mathcal{D}g \mathcal{D}b \mathcal{D}\bar{\psi} \mathcal{D}\psi \\ & \times \exp \left\{ - \int_{M_3} \operatorname{Tr} \left[\frac{k}{2\pi} (-ig^{-1} \partial_\tau g) \partial_t (ig^{-1} \partial_x g) - \frac{1}{2} b^2 + b \left(-ig^{-1} \partial_\tau g - \frac{k}{2\pi} \partial_t (ig^{-1} \partial_x g) \right) + \bar{\psi} \Delta \psi \right] dx dt d\tau \right\}, \end{aligned} \quad (27)$$

where we have used eq. (22). Our aim is to interpret $-ig^{-1} \partial_x g$ as one of the components of a gauge field A_i in three dimensions. In order to make arise the other A_i components we introduce

$$h(x, t, \tau) = g(x, t, \tau) \exp[i\alpha(x, t, \tau)], \quad (28)$$

with α in the Lie algebra of $SU(N)$, and insert the identity

$$1 = \int \mathcal{D}h \delta(\alpha[h]) \quad (29)$$

in $\mathcal{L}_{\text{BRST}}[S^{(n+1)}]$ given by (27):

$$\mathcal{L}_{\text{BRST}}[S^{(n+1)}] = \int_{M_3} \mathcal{D}g \mathcal{D}h \mathcal{D}b \mathcal{D}\bar{\psi} \mathcal{D}\psi \delta(\alpha[h]) \times \exp \left\{ - \int_{M_3} \text{Tr} \left[\frac{k}{2\pi} h^{-1} \partial_\tau h \partial_t (g^{-1} \partial_x g) - \frac{1}{2} b^2 + b \left(-i h^{-1} \partial_\tau h - \frac{k}{2\pi} \partial_t (i g^{-1} \partial_x g) \right) + \bar{\psi} \Delta \psi \right] dx dt d\tau \right\}. \quad (30)$$

In order to match the dimensions of the stochastic time τ with those of x, t it is natural to introduce a parameter μ with dimensions of mass and to make

$$\tau \rightarrow \mu^{-1} \tau. \quad (31)$$

We now perform the following change of variables:

$$A_x = -i g^{-1} \partial_x g, \quad (32)$$

$$A_\tau = -i h^{-1} \partial_\tau h. \quad (33)$$

Concerning the associated jacobian, one can easily see that [15]

$$\mathcal{D}g = \det^{-1}(D_x) \mathcal{D}A_x, \quad \mathcal{D}h = \det^{-1}(D_\tau) \mathcal{D}A_\tau. \quad (34)$$

Here D_x and D_τ are the covariant derivatives

$$D_x = \partial_x + i[A_x,], \quad D_\tau = \partial_\tau + i[A_\tau,]. \quad (35)$$

With this, Δ takes the form

$$\Delta = \mu D_\tau + \frac{k}{2\pi} \partial_t D_x. \quad (36)$$

Using (28) and (33) we can write

$$\delta(\alpha) = \det(D_x D_\tau) \delta(F_{x\tau}). \quad (37)$$

(It is evident that the condition $F_{x\tau} = 0$ implies $h = g$.) Using all these results and the representation

$$\delta(F_{x\tau}) = \int \mathcal{D}A_t \exp \left(\frac{k}{2\pi} \int_{M_3} \text{Tr}(A_t F_{x\tau}) dx dt d\tau \right), \quad (38)$$

$\mathcal{L}_{\text{BRST}}[S^{(n+1)}]$ becomes

$$\mathcal{L}_{\text{BRST}}[S^{(n+1)}] = \int \mathcal{D}A_x \mathcal{D}A_t \mathcal{D}A_\tau \mathcal{D}b \mathcal{D}\bar{\psi} \mathcal{D}\psi \times \exp \left\{ - \int_{M_3} \text{Tr} \left[-\frac{k}{2\pi} A_\tau \partial_t A_x - \frac{k}{2\pi} A_t F_{x\tau} - \frac{b^2}{2\mu} + \frac{b}{\mu} \left(\mu A_\tau + \frac{k}{2\pi} \partial_t A_x \right) + \frac{1}{\mu} \bar{\psi} \left(\mu D_\tau + \frac{k}{2\pi} \partial_t D_x \right) \psi \right] dx dt d\tau \right\}. \quad (39)$$

It is easy to see that the first two terms appearing in the exponential (39) can be recast in the form of the CS action [5]

$$S_{\text{CS}} = \frac{k}{4\pi} \int_{M_3} \epsilon^{ijk} \text{Tr}(A_i \partial_j A_k + \frac{2}{3} A_i A_j A_k) d^3 y, \quad (40)$$

where k is an integer and the indices run over 0, 1, 2. Indeed, under appropriate boundary conditions (say $A_0=0$ on ∂M_3), S_{CS} can be written in the form

$$S_{CS} = \frac{k}{4\pi} \int_{M_3} \epsilon^{ij} \text{Tr}(-A_i \partial_0 A_j + A_0 F_{ij}) d^3y \quad (41)$$

with $i, j = 1, 2$.

Now, if we relabel variables on M_3 as

$$x \rightarrow y_2, \quad t \rightarrow y_0, \quad \tau \rightarrow y_1, \quad (42)$$

the partition function (39) reads

$$\mathcal{Z}_{\text{BRST}}[S^{(n+1)}] = \int \mathcal{D}A_0 \mathcal{D}A_i \mathcal{D}b \mathcal{D}\bar{\psi} \mathcal{D}\psi \\ \times \exp \left\{ - \int_{M_3} \text{Tr} \left[\frac{k}{4\pi} \epsilon^{ij} (-A_i \partial_0 A_j + A_0 F_{ij}) - \frac{b^2}{2\mu} + \frac{b}{\mu} \left(\mu A_1 + \frac{k}{2\pi} \partial_0 A_2 \right) + \frac{1}{\mu} \bar{\psi} \left(\mu D_1 + \frac{k}{2\pi} \partial_0 D_2 \right) \psi \right] d^3y \right\}, \quad (43)$$

where the first two terms correspond to S_{CS} in the form (41) and the rest are the gauge-fixing and ghost terms associated with the gauge condition

$$\frac{1}{\mu} A_1 - \frac{k}{2\pi} \partial_0 A_2 = 0 \quad (44)$$

(note that the scale μ introduced through τ rescaling (eq. (31)) only affects the BRST exact terms related to the gauge fixing and hence $\mathcal{Z}_{\text{BRST}}$ is μ independent). So we have proved that

$$\mathcal{Z}_{\text{BRST}}[S^{(n+1)}] = \mathcal{Z}_{\text{BRST}}[S_{CS}], \quad (45)$$

and then from eq. (24) we have

$$\mathcal{Z}_{\text{stoch}}^{\text{WZW}} = \mathcal{Z}_{\text{BRST}}[S_{CS}]. \quad (46)$$

Thus, starting from a Wess–Zumino–Witten theory in two dimensions we have been able to establish an identity between its stochastic partition function and the BRST partition function for the Chern–Simons theory in $d=3$. This result encompasses the link found in refs. [3,10–12] following a somehow opposite route, i.e. that leading from CS theory to WZW theory using canonical quantization.

The stochastic quantization approach is particularly adapted to the computation of vacuum expectation values in a simple way. A first interesting application of our result should then be the evaluation of Wilson lines in CS theory using $\mathcal{Z}_{\text{stoch}}^{\text{WZW}}$. Another interesting issue which deserves investigation is the one related to the construction of wave functionals describing physical states for the CS theory. Recall that in the scalar example the wave functional for a model with action $S^{(n+1)}$ was $\Psi = \exp(-iS^{(n)})$; moreover, this result was clearly related to the stochastic quantization of the theory with action $S^{(n)}$. In view of the analogous connection between WZW and CS theories that we have established, one would be tempted to conclude that the WZW action should appear as the phase of a state functional for CS theory. In fact, in order to determine the CS wave functionals one has to choose a polarization (for example, in the $A_0=0$ gauge one sets A_2^g to act as “coordinates” and A_1^g as “momenta”) and then it can be shown [16] that physical states are actually described by functionals $\Psi_{CS}[A_2]$ whose phase (once A_2 is written as $A_2[g]$, see eq. (32)) is the WZW action for g . One has to notice however that in passing from $S^{(n+1)}$ as defined in (25) to the CS action we have introduced constraints in the partition function (30) and hence, within our approach, the connection between $\Psi_{CS}[A_2[g]]$ and $S_{\text{WZW}}[g]$ remains to be investigated more thoroughly. We hope to report on all these issues in a separate work.

We would like to thank Daniel Cabra for helpful discussions. F.A.S. wishes to acknowledge Laurent Baulieu for useful comments that stimulated this investigation. This work was partially supported by CONICET, Argentina.

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